

Assignment 1

Exercise 1.4

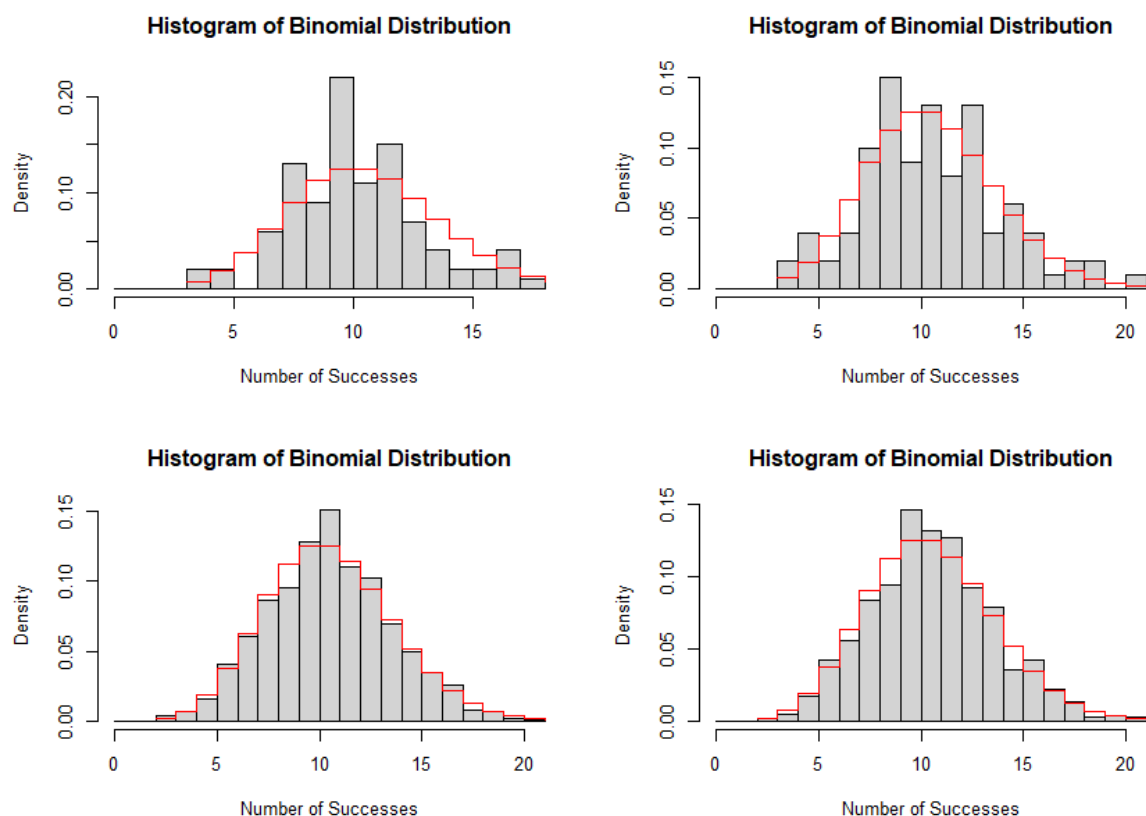
I followed the instructions provided in Assignment 1.

Exercise 1.5

I chose parameter combinations $(n, size, prob) \in \{(100,1000,0.01), (100,10000,0.001), (1000,1000,0.01), (1000,10000,0.001)\}$. The histograms below preserve the order of parameters above.

The parameter *size* increases from left to right in the histograms below. Judging by minor alignment differences between left and right histograms, an increase in the parameter *size* does not seem to improve resemblance of the histogram to the Poisson distribution.

The parameter *n* increases from top to bottom in the histograms below. Judging by observable alignment improvement between top and bottom histograms, an increase in the parameter *n* seems to improve resemblance of the histogram to the Poisson distribution.



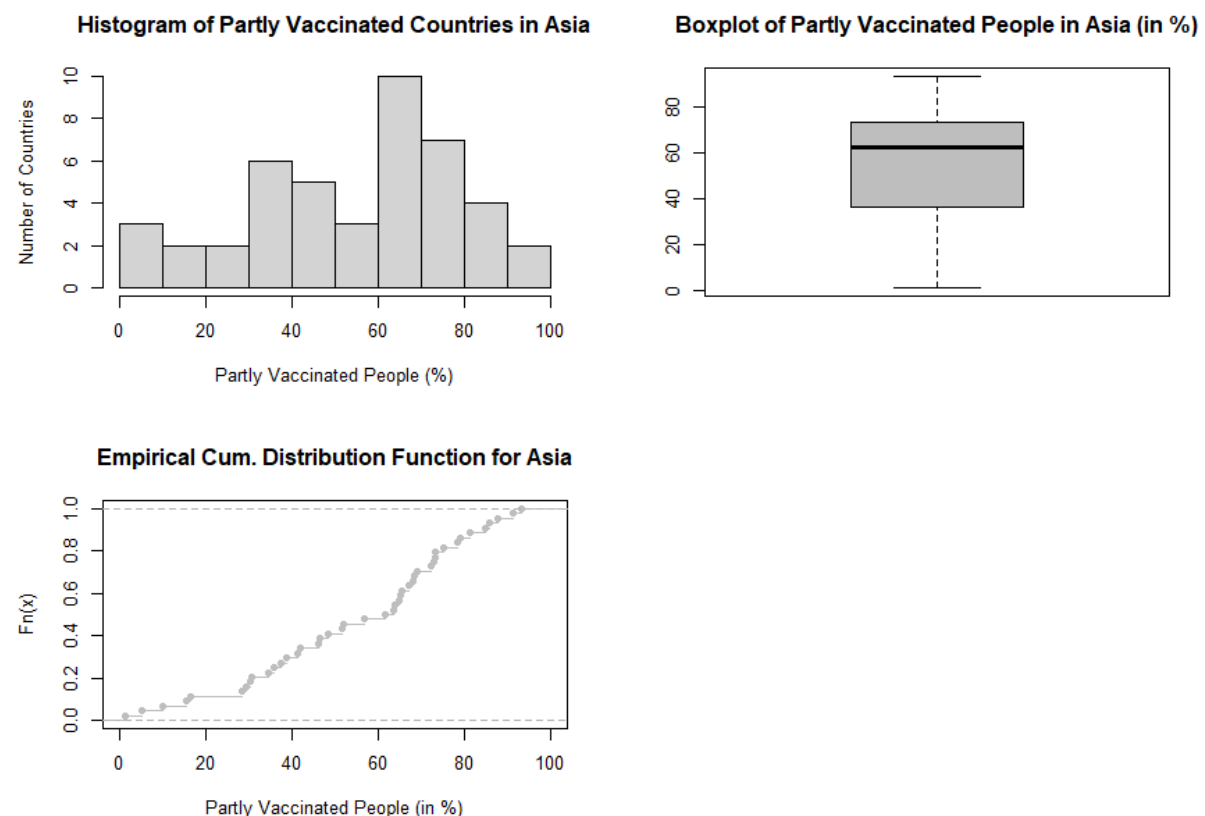
Exercise 1.6

Part a.

I read the column *partly_vacc* from the provided data set, applied functions like *mean()* or *boxplot()*, and obtained the following numerical and graphical summaries.

sample size = 48
 mean = 54.67
 sd = 24.34
 var = 592.32
 min = 1.13
 1st qu. = 36.94
 median = 62.70
 3rd qu. = 73.14
 max = 93.45
 IQR = 36.20
 NA's = 4

The data is not available for 4 out of the 48 observed Asian countries. The 44 observed countries with available data have a mean (partly) vaccination coverage of 54.67% with a standard deviation of 24.34 and a variance of 592.32. The lowest and highest vaccination rates in Asia are 1.13% and 93.45%, respectively. The quartiles indicate that a fourth of the Asian countries with the lowest vaccination coverage has a vaccination coverage lower than 36.94%, a half of the countries has a vaccination coverage lower than 62.70%, and the quarter of Asian countries with the highest vaccination coverage has the vaccination coverage higher than 73.14%. Hence, the difference between the fourth of the countries with highest and lowest vaccination rates is 36.20%.



The histogram above splits the vaccination coverage into 10 bins representing increases of 10% in the partly vaccinated population. For example, it shows that there are 6 countries in Asia that have vaccination coverage in the interval [30%, 40%).
 The box plot above shows the quartiles, median, IQR, minimum and maximum of the vaccination levels in Asian countries.

The eCDF above empirically estimates a probability of a vaccination coverage of an Asian country to be equal or lower than a given value. For example, the probability that an Asian country has a 60% or lower vaccination coverage is approximately 0.5.

Part b.

I read the columns *partly_vacc* and *human_development_index*, and determined the following numerical and graphical summaries.

mean partly vaccination coverage = 54.67

mean HDI = 0.740

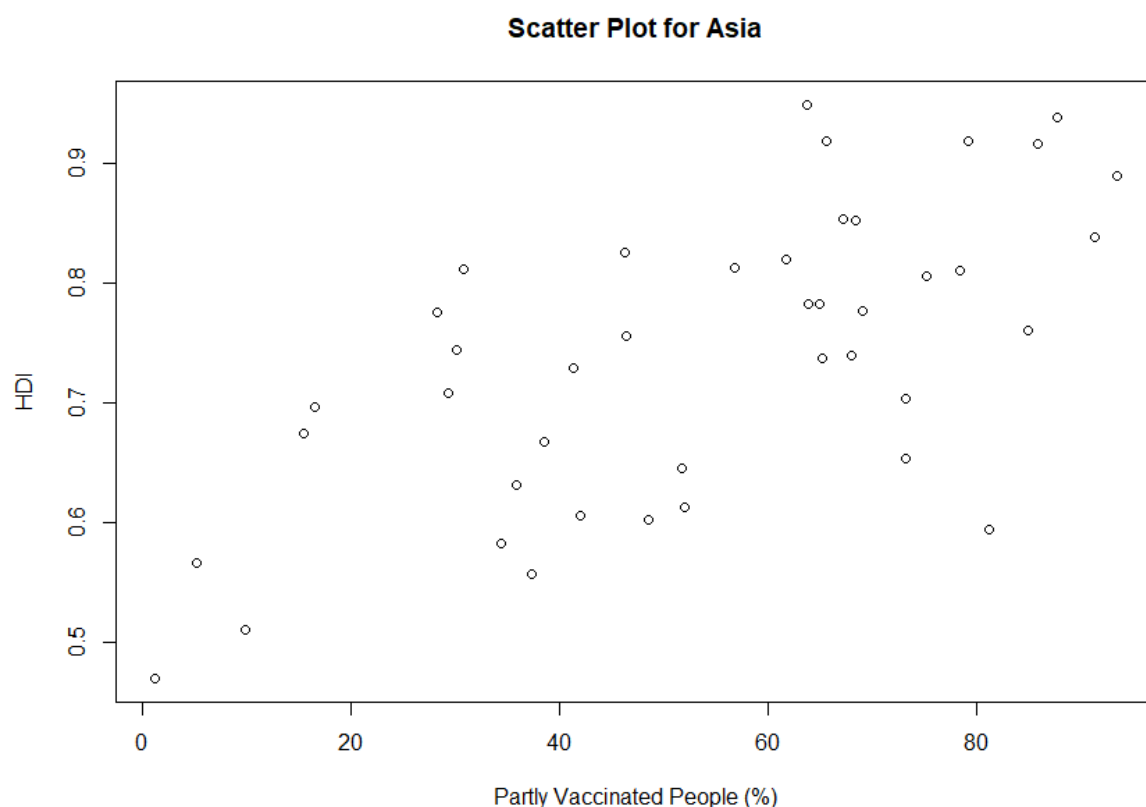
covariance of HDI and vaccination coverage = 1.95

correlation coefficient of HDI and vaccination coverage = 0.65

Spearman's rank correlation coefficient = 0.59

Kendall's rank correlation coefficient = 0.42

The observed Asian countries with available data have an average vaccination coverage of 54.67% and HDI of 0.740. The positive covariance of 1.95 indicates that Asian countries with a higher vaccination coverage tend to have higher HDI, and vice versa. The high values (>0.5) of correlation coefficients, except for the Kendall's, suggest a strong correlation and hence a high degree of linear relation between the vaccination coverage and HDI.



The scatter plot above suggests that countries with a low/high vaccination coverage tend to have low/high HDI.

Appendix

Exercise 1.4

```
lognorm <- function(n, mu, sigma) {  
  set.seed(20220286)  
  
  x = rlnorm(n, meanlog = mu, sdlog = sigma)  
  
  quants = quantile(x, c(0.05, 0.1, 0.25, 0.5, 0.75, 0.9, 1))  
  loc = mean(x)  
  spread = sd(x)  
  stud_no = 2669137  
  
  mylist = c(quants, loc, spread, stud_no)  
  
  save(mylist, file="C:/Users/david/OneDrive/VU/Statistical Data  
Analysis/Assignment1/myfile1_75.RData")  
}
```

Exercise 1.5

```
bin <- function(n, size, prob) {  
  x = rbinom(n, size, prob)  
  u = seq(min(x), max(x)+1, 1)  
  v = dpois(u, size*prob)  
  
  hist(x, prob=T, breaks=0:(max(x)+1)-0.001,  
       xlab = "Number of Successes", main = "Histogram of Binomial Distribution")  
  lines(u, v, type="s", col="red")  
}
```

Exercise 1.6

Part a.

```
covid_data = read.csv("C:/Users/david/OneDrive/VU/Statistical Data Analysis/Assignment  
1/owid_covid_data_selection.csv", header=TRUE)  
covid_data_asia = covid_data[covid_data$continent == 'Asia',]  
partly_vac = covid_data_asia$partly_vacc  
  
length(partly_vac)  
summary(partly_vac)  
sd(na.omit(partly_vac))  
var(na.omit(partly_vac))  
IQR(na.omit(partly_vac))  
  
par(mfrow=c(2,2))
```

```
hist(partly_vac, xlab = "Partly Vaccinated People (%)",  
     ylab = "Number of Countries",  
     main = "Histogram of Partly Vaccinated Countries in Asia")
```

```
boxplot(partly_vac, main="Boxplot of Partly Vaccinated People in Asia (in %)", col=c("grey",2))
```

```
plot(ecdf(covid_data_asia$partly_vacc), col="grey", main="Empirical Cum. Distribution Function for  
Asia", xlab="Partly Vaccinated People (in %)", xlim=c(0,100))
```

Part b.

```
bivariate <- cbind(covid_data_asia$partly_vacc, covid_data_asia$human_development_index)
```

```
colMeans(bivariate, na.rm=T)
```

```
bivariate_woNA <- bivariate[-which(is.na(covid_data_asia$human_development_index)  
                                | is.na(covid_data_asia$partly_vacc)), ]
```

```
cov(bivariate_woNA)
```

```
cor(bivariate_woNA)
```

```
cor(bivariate_woNA, method="spearman")
```

```
cor(bivariate_woNA, method="kendall")
```

```
plot(bivariate_woNA, xlab = "Partly Vaccinated People (%)",  
     ylab = "HDI", main = "Scatter Plot for Asia")
```